Using (A14)–(A16), one finds the standard quadratic identity

$$T^{A}{}_{BC}T_{A}{}^{BC} = \frac{2}{3}T_{\mu}T^{\mu} + \frac{1}{24}S_{\mu}S^{\mu} + \tilde{q}_{\lambda\mu\nu}\tilde{q}^{\lambda\mu\nu}, \tag{A20}$$

where we have denoted $\tilde{q} \equiv P^{(q)}T$ for the *standard* normalization of the traceless piece.

Normalization used in the main text.

For later convenience—and to match the coefficient choice used in Section 4—we rescale the traceless irrep by a constant factor and *define*

$$q_{\lambda\mu\nu} \equiv \frac{1}{\sqrt{2}}\tilde{q}_{\lambda\mu\nu}, \qquad \Longrightarrow \qquad T^{A}{}_{BC}T_{A}{}^{BC} = \frac{2}{3}T_{\mu}T^{\mu} + \frac{1}{24}S_{\mu}S^{\mu} + 2\,q_{\lambda\mu\nu}q^{\lambda\mu\nu}. \tag{A21}$$

The projector formulas (A14)–(A16) are unchanged; only the bookkeeping name "q" for the traceless image carries the fixed $\sqrt{2}$ factor. (This harmless convention matches the positivity split used in the main text and simplifies numerical diagnostics; no physics depends on it.) After applying $\Pi_{\mathrm{PT}}[\cdot]$ to either side of (A21), the scalar is manifestly real (Theorem 1).

Appendix B.3. Compatibility with the Scalar Projector

The projectors $P^{(X)}$ are algebraic and commute with $\Pi_{\mathrm{PT}}[\cdot]$ at the scalar level: for any two torsions T, T',

$$\Pi_{\mathrm{PT}}\big[(P^{(X)}T)_{\lambda\mu\nu}\,(P^{(Y)}T')^{\lambda\mu\nu}\big] = \delta^{XY}\,\Pi_{\mathrm{PT}}\big[(P^{(X)}T)_{\lambda\mu\nu}\,(P^{(X)}T')^{\lambda\mu\nu}\big]. \tag{A22}$$

Moreover, the mixed scalar $T_{\mu}S^{\mu}$ is PT-odd and is annihilated by $\Pi_{\mathrm{PT}}[\cdot]$ (Theorem 1). Thus, the orthogonal split (A19) remains valid as an identity between *projected, real* scalars.

Appendix B.4. Single-Vector No-Go for the Traceless Irrep

We now formalize the statement invoked in Section 4.1.

Proposition A3 (single-vector no-go)**.** *Let v_{μ} be any nonzero covector. There is no nonvanishing tensor of the form $q_{\lambda\mu\nu}(v)$, linear in v_{μ}, that (i) is antisymmetric in $\mu\leftrightarrow\nu$, (ii) obeys $q^{\nu}{}_{\mu\nu} = q_{\lambda\mu}{}^{\mu} = 0$, and (iii) satisfies $\epsilon^{\mu\nu\rho\sigma}q_{\nu\rho\sigma} = 0$. Equivalently, the traceless irrep cannot be constructed from a single vector.*

Proof. The most general Lorentz-covariant tensor built linearly from a single v_{μ} and antisymmetric in its last two indices is a linear combination of the two rank-one seeds

$$Y^{A}{}_{BC}(v) = \alpha\,\delta^{A}{}_{[B}v_{C]} + \beta\,\epsilon^{A}{}_{BCD}v^{D}, \tag{A23}$$

with real α, β. Compute its traces and axial contraction:

$$Y^{\nu}{}_{\mu\nu}(v) = \alpha\,\delta^{\nu}{}_{[\mu}v_{\nu]} = \frac{\alpha}{2}\left(\delta^{\nu}{}_{\mu}v_{\nu} - \delta^{\nu}{}_{\nu}v_{\mu}\right) = -\frac{3\alpha}{2}\,v_{\mu}, \tag{A24}$$

$$Y_{\lambda\mu}{}^{\mu}(v) = \alpha\,\delta_{\lambda[\mu}v_{\nu]}g^{\mu\nu} = 0, \tag{A25}$$

$$\epsilon^{\mu\nu\rho\sigma}Y_{\nu\rho\sigma}(v) = \beta\,\epsilon^{\mu\nu\rho\sigma}\epsilon_{\nu\rho\sigma D}v^{D} = -\,3!\,\beta\,\delta^{\mu}{}_{D}v^{D} = -6\beta\,v^{\mu}, \tag{A26}$$

where we used $\epsilon^{\mu\nu\rho\sigma}\epsilon_{\alpha\nu\rho\sigma} = -3!\,\delta^{\mu}{}_{\alpha}$ and $d = 4$. Requiring the trace constraints $Y^{\nu}{}_{\mu\nu} = Y_{\lambda\mu}{}^{\mu} = 0$ forces $\alpha = 0$, and the axial constraint forces $\beta = 0$. Therefore, the only admissible linear combination is the trivial one, $Y^{A}{}_{BC} \equiv 0$, proving the claim. □